

AUTONOMOUS TRADING SYSTEM

Second Amended Submission to the Court of Claude

Petitioner Corbin	System TradingAgents / Alpaca	Re Resubmission after amended draft. Pre-empts the three sealed stages.
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I. Posture of This Submission

The Grand Jury affirmed the architecture (8/10). The Prosecution executed the first draft on one finding: **paper trading does not validate live edge**. The amended draft adopted that finding and rebuilt the validation path around it. This submission does not re-argue the amended draft. It assumes the amended draft already clears the Grand Jury and the Prosecution, and it is written to survive the three stages that were never reached — the Analysts, the Tribunal, and the Executioner.

Those stages attack different things. The Analysts attack the numbers: *where is the edge, and is it statistically real?* The Tribunal attacks under adversarial contact: *does the edge survive costs, regimes, and the system's own machine-generated reasoning?* The Executioner attacks operations: *does the system kill the operator before the market does?* Each section below names the attack it is built to absorb.

One concession governs everything that follows. **A validation gauntlet with no stated edge filters to cash forever.** The amended draft built an immaculate filter and never named what it was filtering for. That omission is the single largest opening the Analysts have. This submission closes it first.

II. The Edge Thesis — What Is Actually Mispriced

Anticipates: The Analysts ("a filter is not an alpha").

Edge is not claimed as a vibe. It is claimed as four narrow, economically-motivated sleeves, each with a documented reason the mispricing exists and a reason it survives being known. The system does not need all four to work. It needs the gauntlet to find which, if any, survive contact with real fills.

Sleeve	Edge hypothesis	Why it persists (not arbitrated away)	Why a small long-only book can capture it
Pullback-support	A controlled retracement inside intact leadership mean-reverts when the move is non-fundamental.	Short-horizon reversal is a liquidity/inventory premium, not free money: it requires patiently absorbing supply. Collin-Dufresne & Daniel estimate ~10% of idiosyncratic shocks are temporary with a ~2.5-day half-life. Career-risk and turnover costs keep large players from harvesting it cleanly.	Limit orders provide, not take, liquidity — the premium is paid to the patient buyer. Small size avoids the capacity wall that erases it at scale.
Breakout-continuation	Medium-horizon strength persists when stock, industry, and price-level context align.	Jegadeesh & Titman (3–12M momentum), Moskowitz & Grinblatt (industry momentum), George & Hwang (52-week-high anchoring). Underreaction + disposition effect are behavioral and durable; the premium compensates for crash risk (Daniel & Moskowitz), which deters levered arbitrage.	Crash risk is bounded by long-only exposure, regime gating, and small size — the system declines to hold the tail that makes momentum dangerous for levered books.
Catalyst-continuation	A fresh positive catalyst continues when event-day gains are retained and peers confirm.	Post-event drift reflects gradual information diffusion and limited attention. It persists because the drift is noisy at the single-name level and uneconomic for large desks to chase one name at a time.	A patient limit buyer on the post-event hold/reclaim captures diffusion without paying the impulse spread.

Sleeve	Edge hypothesis	Why it persists (not arbitrated away)	Why a small long-only book can capture it
Underreaction / event-drift	Public information is absorbed gradually; a muted initial reaction can drift further.	SEC filings (Form 4 insider clusters, 8-K) are public but unevenly parsed; PEAD-style drift is one of the most replicated anomalies precisely because it is administratively costly to exploit name-by-name.	Free, fast, legally-clean SEC EDGAR data (sub-second submissions feed) lets a small book act on filing freshness the crowd is slow to price.

A fifth state, **cash-default**, is not a sleeve. It is the policy fallback whenever no sleeve passes its gates or the evidence is ambiguous, stale, or contradictory. Cash is a first-class output, never a failure.

III. The Benchmark Bar — Positive Is Not the Standard

Anticipates: The Analysts ("you beat zero, not the index").

A long-only equity sleeve that returns less than buy-and-hold on a risk-adjusted, after-cost, after-tax basis is not an edge — it is a worse index fund. Every sleeve is therefore judged against an explicit benchmark, not against zero.

- **Benchmark of record.** Each sleeve declares its benchmark before testing (default: QQQ for mega-cap/liquid names, or a beta-matched SPY position). The metric is realized alpha and information ratio versus that benchmark, not raw PnL.
- **Beta is not alpha.** Returns are decomposed; market beta is subtracted. A sleeve that is merely long beta in a bull tape scores zero edge regardless of headline return.
- **After-tax, after-friction.** Short holding periods are tax-inefficient. The bar is net of modeled short-term tax drag and the conservative cost model of Section VII, so the system is not rewarded for churn that the IRS and the spread erase.
- **Cash has a hurdle too.** Capital sitting in cash is benchmarked against the risk-free rate, so 'hold cash' is honestly priced rather than treated as costless safety.

IV. The Evidence Pyramid, Re-Based

The amended draft's five validation layers are preserved. Two layers are added *beneath* them, because the old pyramid silently assumed an edge existed before it began testing friction. Promotion requires passing every layer in sequence; any layer can demote a sleeve.

Layer	What it validates	Failure mode it isolates
0. Alpha existence	An economically-motivated hypothesis with a persistence argument exists and is pre-registered <i>before</i> data is touched.	Story-free curve-fitting; building a filter with no candidate edge.
1. Statistical pre-validation	A statistical edge exists ex-ante on walk-forward OOS data, net of multiple-testing correction.	Overfitting and data mining.
2. Benchmark survival	The edge is risk-adjusted alpha versus the declared benchmark, after tax and friction — not just positive.	Beta dressed up as skill.
3. Friction survival	The edge persists net of an adversarially-conservative cost, spread, borrow, and impact model plus a capacity ceiling.	Slippage and market impact.
4. Live-shadow execution	Live-eligible orders are stamped against live quotes, never filled; achievable fills are compared to modeled fills.	Adverse selection and fill quality.
5. Tiny-live confirmation	The only layer that validates true edge. Bounded real capital, real fills, pre-registered gate.	The transfer to real money itself.

Layer	What it validates	Failure mode it isolates
6. Risk-budgeted scale-up	Size grows only as the live sample confirms the backtest within confidence bounds.	Regime drift during deployment.

V. Statistical Honesty: Power, Decay, and the Multiple-Testing Trap

Anticipates: The Analysts ("your sample can never reach significance before the edge dies").

The amended draft's deflated Sharpe handles trials *within* one sleeve. It does not handle three problems the Analysts will raise: statistical power, edge decay, and selection bias *across* sleeves and across time.

- **Power budget, stated honestly.** Each sleeve pre-registers the minimum number of *independent* trades needed for its target effect size at fixed power. A swing sleeve that makes a few bets a month and needs hundreds of independent trades is told, in writing, that confirmation will take years — and is preferred only if its edge is plausibly that durable. Higher-bet-count sleeves (pullback, event-drift) are prioritized precisely because they reach power faster.
- **Decay-aware testing.** The system assumes edges decay. It uses sequential testing with a pre-registered stopping rule and an alpha-spending schedule, so it can confirm or kill faster without inflating false positives — rather than running an open-ended test an edge outlives.
- **Family-wise error across sleeves.** Running many sleeves and re-evaluating continuously is itself a multiple-comparisons machine. A portfolio-level error budget caps the family-wise false-discovery rate across every sleeve and variant ever tested; evaluations happen on a fixed cadence, not whenever a curve looks good.
- **Pre-registration freeze.** Hypothesis, entry/exit rules, target metric, benchmark, and minimum sample are frozen before testing. Changing any of them restarts the clock. This is the mechanism that removes the freedom that produces p-hacking.

VI. The LLM Ablation Gate — Reasoning Must Earn Its Place

Anticipates: The Tribunal ("advisory-only models are unfalsifiable decoration that still hallucinate").

The amended draft made the local models advisory-only, verified, and calibration-scored. The Tribunal's rejoinder is sharp: if the models never size and never trigger, either they add measurable edge or they are decoration that imports hallucination risk for nothing. This submission resolves that by making the LLM layer **prove its lift or be muted**.

- **Ablation gate.** Every sleeve is walk-forward tested twice: deterministic screen alone, and deterministic screen plus the LLM hypothesis layer. The LLM layer is permitted into the live path *only* where it produces statistically significant incremental alpha over the deterministic baseline. Where it does not, it is muted for that sleeve. Hallucination risk is thereby converted from a worry into a measured quantity that must be paid for in edge.
- **Claim-level verification.** Every agent claim carries a machine-checkable source field. A verifier re-checks each claim against its source; unverifiable or contradicted claims are dropped and logged. The model cannot smuggle an unsourced assertion into a decision.
- **Calibration scoring.** Each agent's directional forecasts are Brier-scored over time. Persistently miscalibrated agents are down-weighted or muted automatically.
- **Mandatory abstention and disagreement flag.** Insufficient verified evidence resolves to cash. Material model disagreement demotes the intent to watch and never auto-trades.

- **Determinism at the boundary.** Model confidence never scales position size. Sizing is a deterministic, auditable function of verified features only.

VII. Adversarial Cost Discipline — Breaking the Calibration Circularity

Anticipates: The Tribunal ("you calibrate costs on fills you don't have, so you'll pass non-edges").

Friction survival depends on a cost model, but a faithful cost model needs live fills the system does not yet have. An optimistic model would wave weak sleeves through. The fix is to make the cost model adversarial by default and let reality only ever make it cheaper.

- **Conservative priors.** Until live fills exist, the model assumes the worst defensible case: full half-spread paid, a non-trivial market-impact term scaled by participation, full borrow where relevant, and no price improvement. A sleeve must survive the pessimistic model, not an aspirational one.
- **Costs ratchet one way.** Live fill evidence can only *lower* the assumed cost, and only after a minimum sample. The system can never assume it is getting better fills than it has demonstrated.
- **Capacity ceiling per sleeve.** Each sleeve declares a maximum participation rate and dollar size beyond which its modeled edge is assumed to vanish. Size is capped well under that ceiling in early live.
- **Paper size > live size by design.** Live size starts far below paper size and initial ADV caps are materially tighter, because paper validates the environment, never executable size.

VIII. Regime Permission, Not Regime Reaction

Anticipates: The Tribunal ("your drift kill-switch fires after the loss, by construction").

A detector that pauses a sleeve *after* it diverges is, by definition, lagging. The system cannot pretend otherwise. It does two things instead: it reduces risk *before* the regime turns where the signal is forecastable, and it bounds the loss it is willing to take inside the detection lag.

- **Volatility-managed sizing.** Following Moreira & Muir, the system scales size *down* when volatility rises — volatility is forecastable even when returns are not. Risk comes off ahead of the stress, not after it.
- **Regime as a permission overlay.** Macro/regime state controls which sleeves may act and at what size; it is not a separate alpha story. Momentum-style sleeves are throttled or paused in risk-off and panic states, where Daniel & Moskowitz show the crashes cluster.
- **Bounded-lag loss.** The maximum loss a sleeve can absorb between divergence and detection is pre-computed and pre-sized. The drift kill-switch is the backstop, not the primary control.
- **Overnight gap budget.** Limit and stop logic is never assumed to protect against gaps. A separate overnight exposure budget and gap-risk penalty govern positions held across the close.

IX. Portfolio Truth — Correlated Sleeves Are Hidden Leverage

Anticipates: The Executioner ("four long-only sleeves long the same mega-caps is one undiversified bet").

Four sleeves that independently fall in love with the same handful of liquid leaders are not four bets; they are one concentrated, correlated bet wearing four hats — leverage in disguise. Risk is therefore measured at the portfolio level, across sleeves, not inside each sleeve alone.

- **Exposure decomposition.** Aggregate exposure is decomposed by name, sector, and factor across all sleeves before any order. Two sleeves pointing at the same name net into one capped position.
- **Hard caps that span sleeves.** Per-name, per-sector, and aggregate-beta caps apply to the whole book, not per sleeve. A portfolio-level drawdown circuit breaker halts all live submission on breach.

- **Physical path separation.** The live execution path is physically separate from research and paper paths, so a research bug cannot reach the live account.

X. Operational Survival — The System Must Not Kill the Operator

Anticipates: The Executioner ("a crash mid-order, a duplicate fill, a stale clock, and you are short your own stop").

A correct strategy with broken operations still loses money. The supervisor runs unattended on scheduled jobs against a brokerage API; the failure modes are mechanical, not statistical, and they are enumerated with controls.

- **Idempotent orders.** Every order carries a deterministic client-side idempotency key derived from the intent, so a retry after a crash cannot create a duplicate position.
- **Reconcile-on-start.** Before any action, the supervisor reconciles its own record of positions and open orders against the broker's truth and fails closed on any mismatch.
- **Single-writer lock.** Only one process may submit at a time. Overlapping scheduled runs cannot race each other into the account.
- **Fail-closed on uncertainty.** Stale data, clock skew beyond tolerance, API error, or missing confirmation all resolve to no action, never to a guess.
- **Dead-man switch.** A heartbeat governs live trading; if the supervisor stops checking in, new live submission is disabled and a documented one-command manual kill flattens or freezes the book.
- **Every decision writes a packet.** Including 'no trade'. Each packet carries hashes, versions, gate outcomes, and the data-integrity report, so any fill is reconstructable after the fact.

XI. Failure-Mode Taxonomy — Every Mode, a Control, a Kill Condition

A control without a kill condition is decoration. Each row names both. The first rows are carried from the amended draft; the remainder are new and answer the three sealed stages.

Failure mode	Control	Kill condition
No real edge	Layer-0 pre-registered persistence argument; benchmark survival layer.	No alpha vs benchmark after multiple-testing correction → sleeve never promoted.
Overfitting / data mining	Walk-forward OOS, purged k-fold + embargo, deflated Sharpe, pre-registration.	Deflated Sharpe below threshold rejects the sleeve outright.
Insufficient power / edge decay	Pre-registered power budget; sequential test with alpha-spending.	Power unreachable before plausible decay horizon → sleeve shelved.
Cross-sleeve selection bias	Family-wise error budget; fixed evaluation cadence.	Family-wise FDR breach freezes new promotions.
Slippage / market impact	Adversarially-conservative cost + impact model; capacity ceiling.	Live slippage exceeds modeled tolerance over minimum sample.
Adverse selection	Live-shadow fill analysis; queue-position and fill-rate comparison.	Live fill rate falls below shadow estimate beyond threshold.
LLM hallucination / no lift	Claim verification; calibration scoring; ablation gate.	No incremental alpha over deterministic baseline → LLM layer muted.
Regime change	Volatility-managed sizing; regime permission overlay; bounded-lag loss.	Regime classifier flip auto-pauses momentum sleeves.

Failure mode	Control	Kill condition
Correlated concentration	Cross-sleeve exposure decomposition; per-name/sector/beta caps.	Aggregate cap or drawdown breaker halts all live submission.
Survivorship bias	Point-in-time universe incl. delisted/acquired names; as-was index membership.	Data-integrity gate fails if universe is not point-in-time.
Staleness / look-ahead	As-of timestamps, freshness stamps, embargo windows on every feature.	Stale or future-dated feature blocks the trade.
Operational fault	Idempotency keys; reconcile-on-start; single-writer lock; dead-man switch.	Any reconciliation mismatch or heartbeat loss disables live submission.

XII. Promotion Ladder — A Sleeve Never Skips a State

Any state can demote to Paused. Cash is always a legal terminal state. Two states are added below the amended ladder to enforce Layers 0 and 2.

State	Entry requirement
Hypothesis-registered	Economically-motivated edge + persistence argument pre-registered before data.
Backtest-validated	Passes Section V statistical gates on walk-forward OOS data.
Benchmark-cleared	Risk-adjusted alpha vs declared benchmark, after tax and friction.
Friction-survived	Edge persists net of conservative cost, impact, and capacity ceiling.
Ablation-cleared	LLM layer either proven additive or muted; deterministic core stands alone.
Shadow-confirmed	Live-shadow fills match model within tolerance over minimum sample.
Tiny-live-confirmed	Pre-registered live statistical gate met on bounded real capital.
Risk-budgeted live	Scaled under portfolio exposure budget and cross-sleeve caps.
Paused / killed	Drift, regime break, invalidator, concentration breach, or data-integrity failure.

XIII. What This Submission Concedes

A system that hides its residual risk invites the next prosecution. These are stated plainly.

- **Decay is reduced, not eliminated.** A real historical edge can still decay after deployment. The claim is narrow: the system detects decay quickly and stops, rather than claiming permanence.
- **Confirmation is slow by construction.** Honest power requirements mean some sleeves take a long time to confirm. The system treats that as a cost to be paid, not a corner to be cut, and prefers faster-sampling sleeves where the edge allows.
- **Free data limits breadth and granularity.** The system compensates with conservative capacity ceilings, freshness gating, and source-conflict detection — not by pretending the data is institutional grade. Delisted-name price history from free sources is the weakest link and is sourced and flagged explicitly.
- **Tiny-live remains the irreducible test.** No backtest or shadow run substitutes for real fills. The system validates edge empirically at small size before risking more, and is designed to fail loudly rather than quietly.
- **The LLM layer may earn nothing.** It is entirely possible the ablation gate mutes the models everywhere. The system accepts that outcome; a deterministic policy engine with no surviving alpha sleeves correctly trades nothing.

XIV. Question Presented

The Court is asked to judge whether this submission severs every inference its sealed stages would have executed. It names a falsifiable edge with a persistence argument rather than assuming one. It judges that edge against a benchmark rather than against zero. It states its statistical power and decay honestly and budgets for selection bias across sleeves. It forces its own machine reasoning to prove incremental lift or be muted. It models cost adversarially, manages regime risk before the fact, measures concentration across sleeves, and hardens its operations against the mechanical failures that kill unattended trading systems. Each layer is falsifiable, each carries a kill condition, and the system can always choose cash.

The Court is asked whether this layered, self-killing, benchmark-anchored validation path is sound, where it remains weakest, and what should be preserved, rewritten, or rejected.

your honor.